INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD

CERTIFICATE FOR THE FINAL VERSION OF DISSERTATION

(To be submitted at the time of Final Dissertation Submission)

This is to certify that the Dissertation entitled " **Diabetic Retinopathy Classification using Deep Learning and Ordinal Regression** " being submitted to the Indian Institute of Technology (Indian School of Mines), Dhanbad, by **Mr. Ayush Jha,** Admission No **21JE0206** for the award of the Degree **Integrated Master of Technology** from IIT (ISM), Dhanbad, is a bonafide work carried out by him, in the Department of **Mathematics and Computing**, IIT (ISM), Dhanbad, under my/our supervision and guidance. The dissertation has fulfilled all the requirements as per the regulations of this Institute and, in my/our opinion, has reached the standard needed for submission. The results embodied in this dissertation have not been submitted to any other university or institute for the award of any degree or diploma.

Prof. Neeru Bala

Professor
Department of Mathematics and Computing
Indian Institute of Technology
(Indian School of Mines), Dhanbad

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD

DECLARATION BY THE STUDENT

(To be submitted at the time of Final Dissertation Submission)

I hereby declare that the work which is being presented in this dissertation entitled " **Diabetic Retinopathy Classification using Deep Learning and Ordinal Regression** " in partial fulfilment of the requirements for the award of the degree of Master of **Integrated Master of Technology in Mathematics and Computing** is an authentic record of my own work carried out during the period from **August 1, 2025** to **May 5, 2026** under the supervision of **Prof. Neeru Bala** Department of **Mathematics and Computing** Indian Institute of Technology (ISM) Dhanbad, Jharkhand, India.

I acknowledge that I have read and understood the UGC (Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations, 2018. These Regulations were published in the Indian Official Gazette on 31st July, 2018.

I confirm that this Dissertation has been checked for plagiarism using the online plagiarism checking software provided by the Institute. At the end of the Dissertation, a copy of the summary report demonstrating similarities in content and its potential source (if any) generated online using plagiarism checking software is enclosed. I herewith confirm that the Dissertation has less than 10% similarity according to the plagiarism checking software's report and meets the MoE/UGC Regulations as well as the Institute's rules for plagiarism.

I further declare that no portion of the dissertation or its data will be published without the Institute's or Guide's permission. I have not previously applied for any other degree or award using the topics and findings described in my dissertation.

_____

(Signature of the Student)

Name of the Student: **Ayush Jha**

Admission No.: **21JE0043**

Department : **Mathematics and Computing**

Indian Institute of Technology

(Indian School of Mines), Dhanbad

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD

CERTIFICATE FOR CLASSIFIED DATA

(To be submitted at the time of Final Dissertation Submission)

This is to certify that the Dissertation entitled " **Diabetic Retinopathy Classification using Deep Learning and Ordinal Regression** " being submitted to the Indian Institute of Technology (Indian School of Mines), Dhanbad by **Mr. Ayush Jha** for award of **Integrated Master of Technology in Mathematics and Computing** does not contain any classified information. This work is original and yet not been submitted to any institution or university for the award of any degree.

Signature of Supervisor	Signature of Student
Name: Prof. Ayush Jha Date: 05/05/2026	Name: Ayush Jha Date: 05/05/2026

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD

CERTIFICATE REGARDING ENGLISH CHECKING

(To be submitted at the time of Final Dissertation Submission)

This is to certify that the Dissertation entitled " **Diabetic Retinopathy Classification using Deep Learning and Ordinal Regression** " being submitted to the Indian Institute of Technology (Indian School of Mines), Dhanbad by **Mr. Ayush Jha**, Admission No **21JE0206**, for the award of **Integrated Master of Technology in Mathematics and Computing** has been thoroughly checked for quality of English and logical sequencing of topics.

Itis hereby certified that the standard of English is good and that grammar and typos have been thoroughly checked.

Signature of Supervisor

Name: Prof. Neeru Bala
Date: 05/05/2026

Signature of Student

Name: Ayush Jha
Date: 05/05/2026

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD

COPYRIGHT ANDCONSENT FORM

(To be submitted at the time of Final Dissertation Submission)

To ensure uniformity of treatment among all contributors, other forms may not be substituted for this form, nor may any wording of the form be changed. This form is intended for original material submitted to the IIT (ISM), Dhanbad and must accompany any such material in order to be published by the IIT (ISM). Please read the form carefully and keep acopy for your files.

TITLE OF DISSERTATION: **Diabetic Retinopathy Classification using Deep Learning and Ordinal Regression**

AUTHOR'S NAME & ADDRESS: **Ayush Jha**
Rajharsh Colony, Bhopal, Madhya Pradesh - 462042

COPYRIGHT TRANSFER

1. The undersigned hereby assigns to Indian Institute of Technology (Indian School of Mines), Dhanbad all rights under copyright that may exist in and to: (a) the above Work, including any revised or expanded derivative works submitted to the IIT (ISM) by the undersigned based on the work; and (b) any associated written or multimedia components or other enhancements accompanying the work.

CONSENT AND RELEASE

2. In the event the undersigned makes a presentation based upon the work at a conference hosted or sponsored in whole or in part by the IIT (ISM) Dhanbad, the undersigned, in consideration for his/her participation in the conference, hereby grants the IIT (ISM) the unlimited, worldwide, irrevocable permission to use, distribute, publish, license, exhibit, record, digitize, broadcast, reproduce and archive; in any format or medium, whether now known or hereafter developed: (a) his/her presentation and comments at the conference; (b) any written materials or multimedia files used in connection with his/her presentation; and (c) any recorded interviews of him/her (collectively, the "Presentation"). The permission granted includes the transcription and reproduction of the Presentation for inclusion in products sold or distributed by IIT(ISM) Dhanbad and live or recorded broadcast of the Presentation during or after the conference.
3. In connection with the permission granted in Section 2, the undersigned hereby grants IIT (ISM) Dhanbad the unlimited, worldwide, irrevocable right to use his/her name, picture, likeness, voice and biographical information as part of the advertisement, distribution and sale of products incorporating the Work or Presentation, and releases IIT (ISM) Dhanbad from any claim based on right of privacy or publicity.
4. The undersigned hereby warrants that the Work and Presentation (collectively, the "Materials") are original and that he/she is the author of the Materials. To the extent the Materials incorporate text passages, figures, data or other material from the works of others, the undersigned has obtained any necessary permissions. Where necessary, the undersigned has obtained all third party permissions and consents to grant the license above and has provided copies of such permissions and consents to IIT (ISM) Dhanbad.

GENERAL TERMS

* The undersigned represents that he has the power and authority to make and execute this assignment.
* The undersigned agrees to indemnify and hold harmless the IIT (ISM) Dhanbad from any damage or expense that may arise in the event of a breach of any of the warranties set forth above.
* In the event the above work is not accepted and published by the IIT (ISM) Dhanbad or is withdrawn by the author(s) before acceptance by the IIT(ISM) Dhanbad, the foregoing copyright transfer shall become null and void and all materials embodying the Work submitted to the IIT(ISM) Dhanbad will be destroyed.
* For jointly authored Works, all joint authors should sign, or one of the authors should sign as authorized agent for the others.

Signature of theAuthor